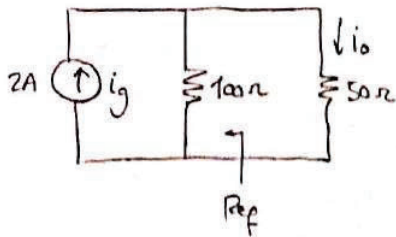


QUESTION 1 a)

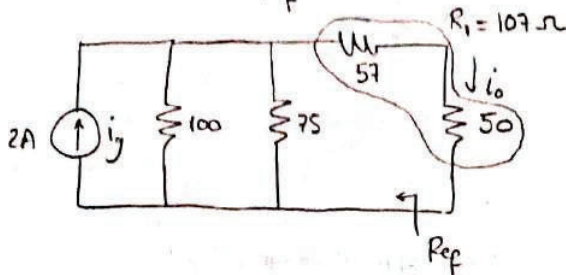
i)



$$i_o = 2 \times \frac{100 \parallel 50}{50} = 1.333 \text{ A}$$

↑
current divider

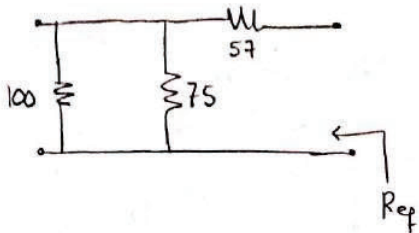
ii)



$$i_o = 2 \times \frac{100 \parallel 75 \parallel 107}{107} = 2 \times \frac{30.6}{107} = 0.572 \text{ A}$$

↑
current divider

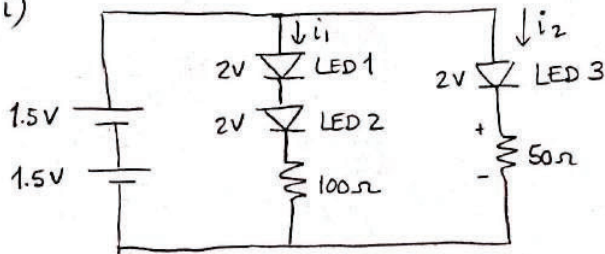
iii)



$$R_{eq} = 57 + (75 \parallel 100) = 99.86 \approx 100 \Omega$$

QUESTION 1 b)

i)



LED1 and LED2 are OFF

because there is not enough voltage to make the 2 of them work - we need 4V to make them work

$$\text{For LED3} \Rightarrow i_2 = \frac{3-2}{50} = 0.02 \text{ A} < i_{max}$$

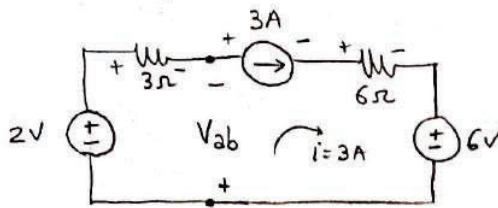
so LED3 is ON

ii)

$$P_{50\Omega} = i_2^2 \times 50 = 0.02^2 \times 50 = 0.02 \text{ W}$$

QUESTION 2a)

i)



$$-2 + 3i - V_{ab} = 0 \Rightarrow \boxed{V_{ab} = -2 + 3i = -2 + 3 \times 3 = 7V}$$

ii)

$$\boxed{P_{2V} = -V \times i = -2 \times 3 = -6W}$$

$$\boxed{P_{3\Omega} = i^2 \times 3 = 3^2 \times 3 = 27W}$$

$$\boxed{P_{3A} = V_{3A} \times 3 = -31 \times 3 = -93W}$$

→ KVL to find V_{3A} :

$$-2 + 3i + V_{3A} + 6i + 6 = 0$$

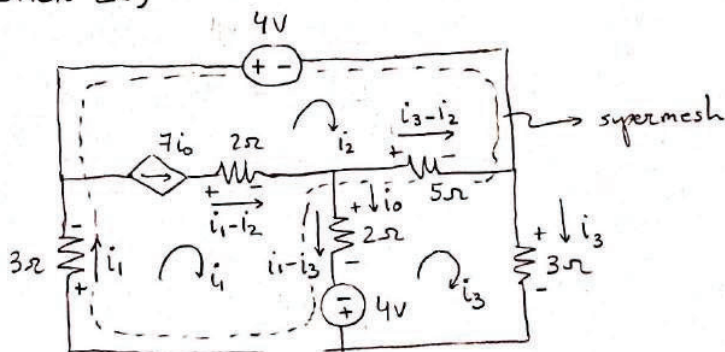
$$V_{3A} = -4 - 9i = -4 - 9 \times 3 = -31V$$

$$\boxed{P_{6\Omega} = i^2 \times 6 = 3^2 \times 6 = 54W}$$

$$\boxed{P_{6V} = +V \times i = +6 \times 3 = 18W}$$

$$\underline{\underline{\sum P = -6 + 27 - 93 + 54 + 18 = 0}}$$

QUESTION 2b)



Mesh equations:

$$\begin{cases} 5i_1 + 5i_2 - 7i_3 = 0 \\ -2i_1 - 5i_2 + 10i_3 = 0 \\ -6i_1 - i_2 + 7i_3 = 0 \end{cases}$$

$$\text{KVL @ supermesh: } 3i_1 + \cancel{4} - 5(i_3 - i_2) + 2(i_1 - i_3) - \cancel{4} = 0$$

$$\rightarrow 5i_1 + 5i_2 - 7i_3 = 0$$

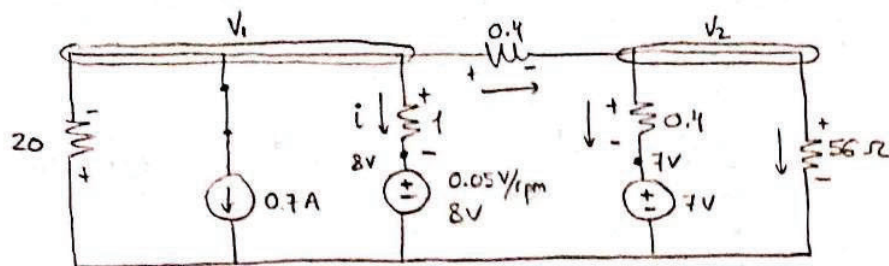
$$\text{KCL @ mesh 3: } 4 - 2(i_1 - i_3) + 5(i_3 - i_2) + 3i_3 = 0$$

$$\rightarrow -2i_1 - 5i_2 + 10i_3 = -4$$

$$\text{Constraint equation: } i_1 - i_2 = 7i_0 \text{ where } i_0 = i_1 - i_3$$

$$\rightarrow i_1 - i_2 = 7(i_1 - i_3); i_1 - i_2 - 7i_1 + 7i_3 = 0 \Rightarrow -6i_1 - i_2 + 7i_3 = 0$$

QUESTION 3



2)

KCL @ node 1: $\frac{0 - V_1}{20} - 0.7 - \frac{V_1 - 8}{1} - \frac{V_1 - V_2}{0.4} = 0$ (1)

KCL @ node 2: $\frac{V_1 - V_2}{0.4} - \frac{V_2 - 7}{0.4} - \frac{V_2}{56} = 0$ (2)

Simplify and solve equations:

$$-\frac{V_1}{20} - V_1 - \frac{V_1}{0.4} + \frac{V_2}{0.4} = 0.7 - 8$$

$$-\left(\frac{1}{20} + 1 + \frac{1}{0.4}\right)V_1 + 2.5V_2 = -7.3 \Rightarrow -3.55V_1 + 2.5V_2 = -7.3 \quad (1)$$

$$\frac{V_1}{0.4} - \frac{V_2}{0.4} - \frac{V_2}{0.4} + \frac{7}{0.4} - \frac{V_2}{56} = 0$$

$$2.5V_1 - \left(\frac{2}{0.4} + \frac{1}{56}\right)V_2 = -17.5 \Rightarrow 2.5V_1 - 5.02V_2 = -17.5 \quad (2)$$

$$\begin{array}{rcl} -3.55V_1 + 2.5V_2 = -7.3 & \begin{array}{l} (\times 5.02) \\ (\times 2.5) \end{array} & \begin{array}{l} -17.821V_1 + 12.55V_2 = -36.65 \\ 6.25V_1 - 12.55V_2 = -43.75 \end{array} \\ 2.5V_1 - 5.02V_2 = -17.5 & & \end{array}$$

$$+11.57V_1 = +80.4 \Rightarrow V_1 = \frac{80.4}{11.57} = 6.95V$$

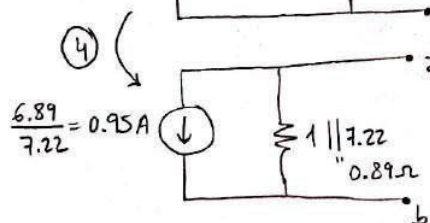
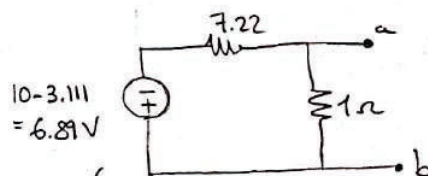
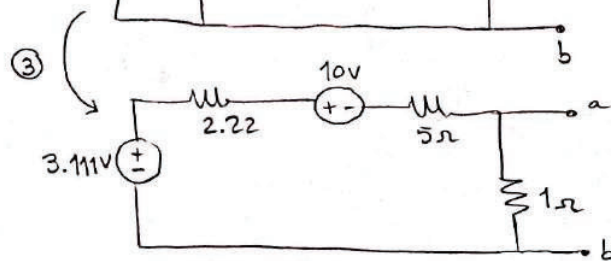
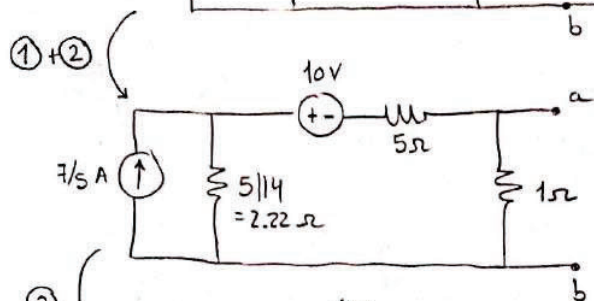
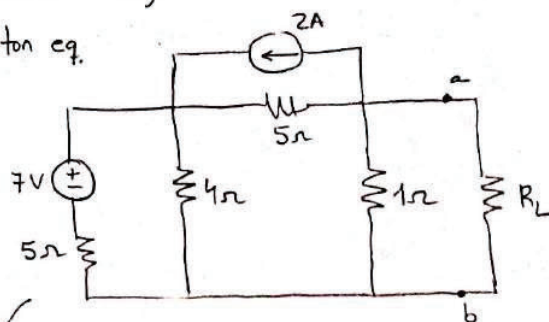
We only need V_1 to calculate i :

$$i = \frac{V_1 - 8}{1} = 6.95 - 8 = -1.05A$$

b) $P_{20\Omega} = (0 - V_1)^2 / 20 = \frac{6.95^2}{20} = 2.415W$

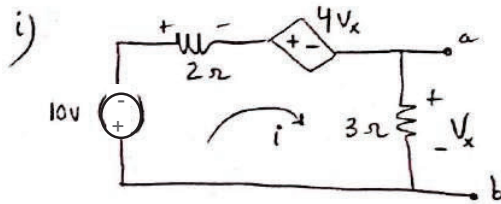
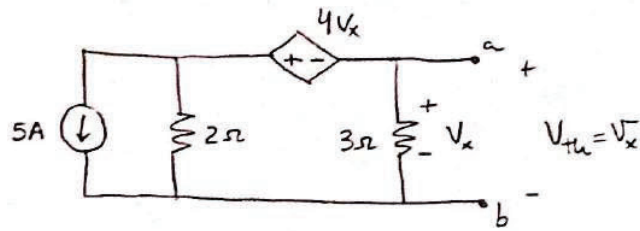
QUESTION 42)

Norton eq.



$$\Rightarrow \begin{cases} I_N = 0.95A \\ R_N = R_{Th} = 0.89\Omega \end{cases}$$

QUESTION 4b



KVL @ mesh: $+10 + 2i + 4V_x + V_x = 0$, where $V_x = V_{th} = 3i$

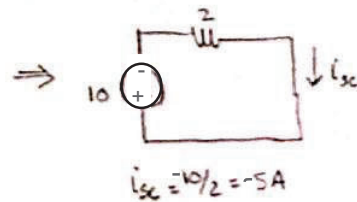
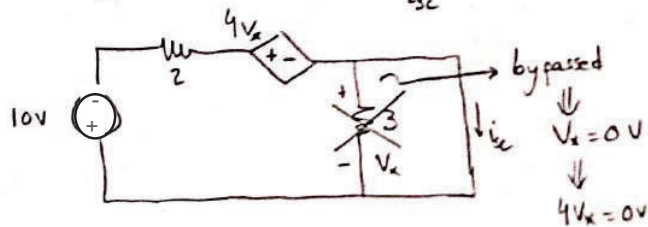
$$+10 + 2i + 12i + 3i = 0$$

$$+10 + 17i = 0 \Rightarrow i = \frac{-10}{17} = -0.59 \text{ A}$$

$$\boxed{V_{th} = V_x = 3i = -1.76 \text{ V}}$$

ii) $R_L = R_{th}$

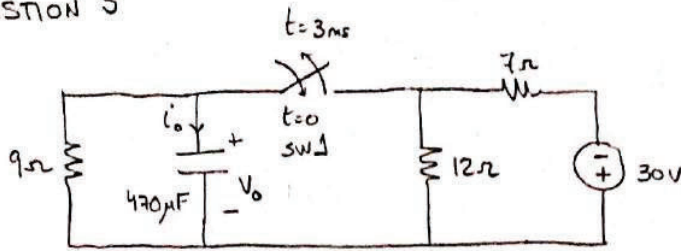
Using method 2: $R_{th} = \frac{V_{th}}{I_{sc}}$



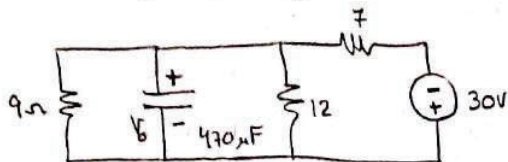
$$\boxed{R_{th} = \frac{-1.76}{-5} = 0.35 \Omega}$$

iii) $\boxed{P_{max} = \frac{V_{th}^2}{4R_{th}} = \frac{1.76^2}{4 \times 0.35} = 2.21 \text{ W}}$

QUESTION 5



- a) V_o for $0 \leq t < 3 \text{ ms}$
(switch closed)

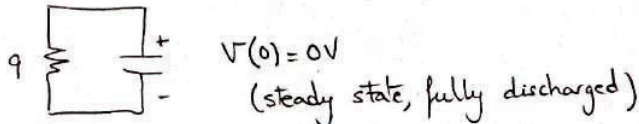


Step response

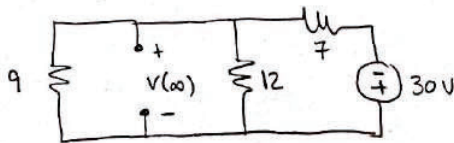
$$V_o = V(\infty) + [V(0) - V(\infty)] e^{-t/\tau}$$

where $\tau = RC$

- * Calculate $V(0)$ for $t=0$
(switch open)



- * Calculate $V(\infty)$ for $0 \leq t < 3 \text{ ms}$



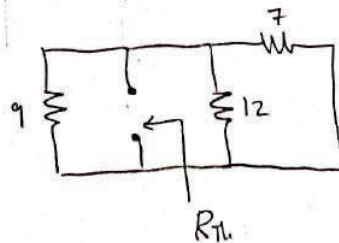
$$V(\infty) = -30 \frac{12 \parallel 9}{12 \parallel 9 + 7} = -30 \frac{5.14}{12.14} = -12.7 \text{ V}$$

↑
voltage divider

- * Calculate τ for $0 \leq t < 3 \text{ ms}$

where $\tau = R_{th} \times C = 2.96 \times 470 \mu\text{F} = 1.39 \text{ ms}$

→ Calculate R_{th} :

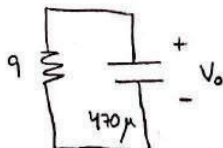


$$R_{th} = 12 \parallel 9 \parallel 7 = 2.96 \Omega$$

* Substitute:

$$V_o(t) = -12.7 (1 - e^{-t/0.00139}) \text{ V}$$

- b) V_o for $t \geq 3 \text{ ms}$:



Natural response

$$V_o(t) = V(3\text{ms}) e^{-(t-0.003)/\tau}$$

where $\tau = RC$

* $V(3\text{ms}) = -12.7 (1 - e^{-0.003/0.00139}) = -11.23 \text{ V}$
(from a)

* $\tau = RC = 9 \times 470 \mu\text{F} = 4.23 \text{ ms}$

* Substitute: $V_o(t) = -11.23 e^{-(t-0.003)/0.00423} \text{ V}$